

EXTRACTION AND VERIFICATION OF INFORMATION FROM SEMI-CATEGORISED DATA

Goma E^{*1}, Bhavana N^{*2}, Divya G^{*3}, Harini N^{*4}

^{*1,2,3,4}Department Of Computer Science Adhiyamaan College Of Engineering, Hosur, India.

ABSTRACT

In modern recruitment processes, the verification is conducted manually, leading to inefficiencies and potential errors. The proposed solution utilizes by automating the verification of biodata applications and supporting documents by Intelligent Document Processing (IDP) techniques, Natural Language Processing (NLP), to streamline the extraction and validation of information from various documents such as educational certificates and GATE scorecards, which may be in image or PDF formats and in languages other than Hindi or English. Candidates will upload their biodata and documents through a web portal, where a Translation API will assist with language discrepancies. Optical Character Recognition (OCR) will be employed to extract text from these documents, storing the information temporarily for validation. NLP techniques will then process this data to ensure accuracy, while MapReduce models will facilitate rapid processing within three seconds per document. In cases of mismatches, alerts will notify users immediately. Upon successful verification, both the documents and validated user data will be securely stored in a database.

Keywords: Verification Systems, Data Accuracy, Security, Random Forest Classifier, Translation, OCR, NLP, Extracted Data, Mapreduce, SSL/TLS, AES Encryption, Data Privacy.

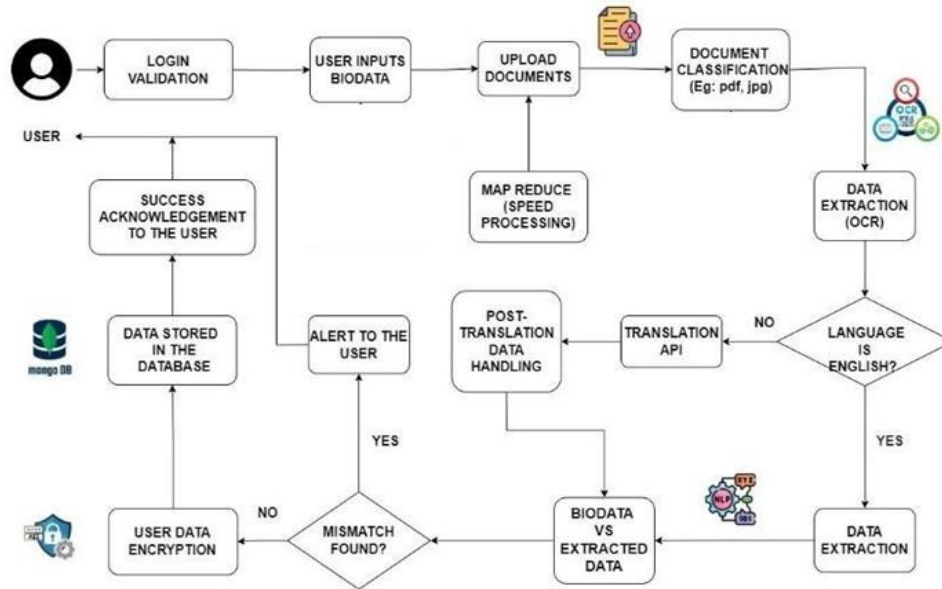
I. INTRODUCTION

In today's digital era, automated document verification plays a crucial role in ensuring data integrity and security in various domains, including government recruitment, financial institutions, and legal services. Manual verification processes are often time-consuming, error-prone, and inefficient, necessitating the adoption of intelligent document processing (IDP) techniques. This research introduces an advanced biodata verification system that automates the process of validating user submitted documents while ensuring accuracy, security, and efficiency. Users submit their biodata along with supporting documents in image or PDF format through a web portal designed for verification purposes. The system employs a Random Forest Classifier to classify file types, ensuring compatibility with the verification process. In cases where documents contain text in regional languages, a Translation API is integrated to convert the text into a standard language for further processing. Optical Character Recognition (OCR) is then used to extract relevant information from the documents, converting scanned text into machine-readable format.

II. LITERATURE SURVEY

Automated verification of candidate documents is essential in modern recruitment systems, as manual checking is slow and prone to errors. This project introduces an Intelligent Document Processing (IDP)-based system that uses AI, OCR, Machine Learning, and Natural Language Processing to analyse and validate biodata and supporting documents. The system can handle scanned, handwritten, and multilingual documents by integrating a Translation API for accurate text extraction. OCR converts uploaded documents into machine-readable text, enabling automated comparison with the candidate's submitted information. NLP-based matching techniques verify important fields such as name, date of birth, certificate numbers, and category details. Distributed processing through MapReduce ensures that verification is completed within a few seconds while maintaining high accuracy. The system also detects mismatches and alerts both candidates and recruiters instantly. Verified documents and extracted data are stored securely for future reference. By automating the entire workflow, the system increases speed, reliability, and transparency in recruitment processes, making the selection procedure more efficient and trustworthy.

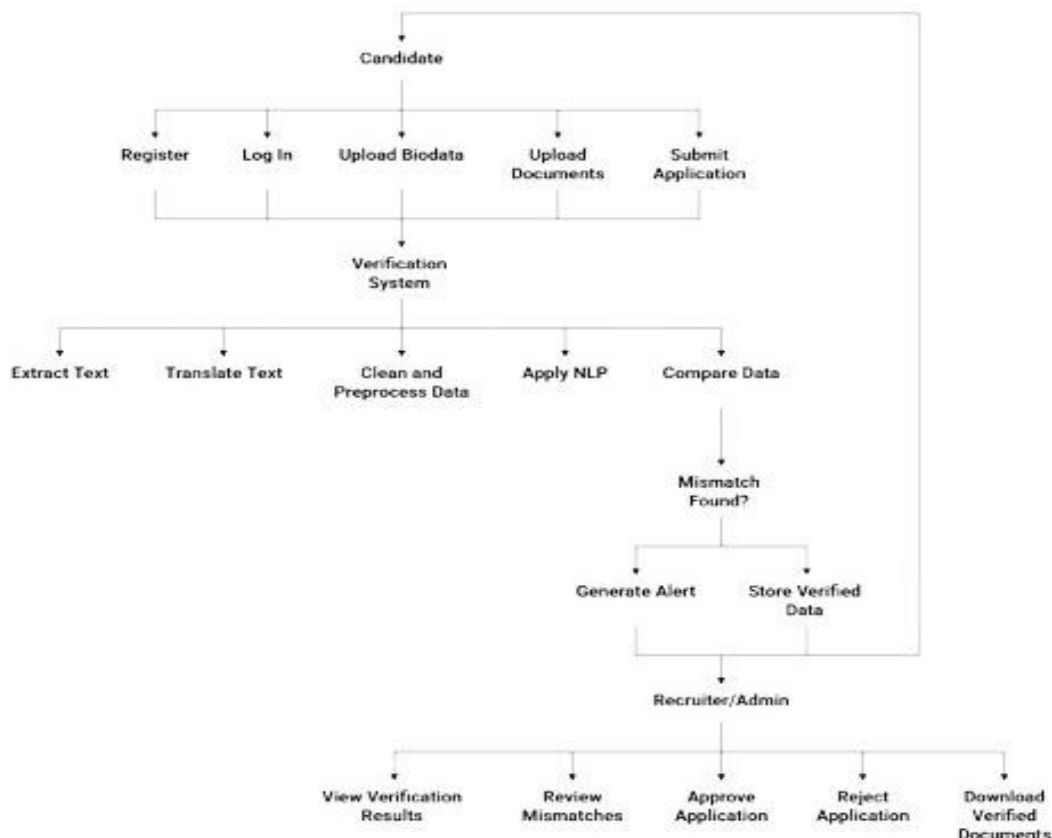
III. ARCHITECTURE DESIGN



EXISTING SYSTEM

- Basic online verification portals exist but lack advanced features such as automated mismatch detection, multilingual translation, and intelligent comparison between extracted text and user-entered data.
- Existing IDP tools are mostly designed for corporate workflows and do not provide a complete end-to-end verification pipeline tailored for government recruitment, especially one that ensures fast processing, secure storage, and instant alerts for inconsistencies.

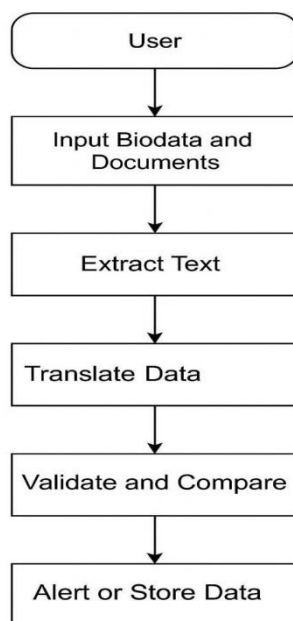
USE CASE DIAGRAM:



IV. PROPOSED SYSTEM

- The proposed system is an Intelligent Document Processing (IDP)–based solution designed to automate the verification of candidate documents using OCR, Machine Learning, NLP, and translation techniques for high-accuracy text extraction and validation.
- The system enables candidates to upload biodata and supporting documents, where OCR extracts text, multilingual content is translated, and NLP algorithms compare the extracted data with user-submitted information to detect inconsistencies or mismatches.
- By employing a MapReduce-based processing model, the system verifies documents within a few seconds while ensuring scalability and efficient handling of large volumes of recruitment data, supported by secure storage and encrypted data transmission.
- The primary goal is to streamline recruitment workflows by reducing manual verification effort, improving accuracy, and ensuring fairness and transparency, thereby supporting faster and more reliable candidate assessment.

DATAFLOW DIAGRAM:



V. MODULE DESCRIPTION

1. User Interface Module:

The User Interface module allows candidates to register, log in, upload biodata, and submit their documents through a clean and accessible web portal designed for ease of use. It provides real-time status messages, progress indicators, and prompts that guide users through the document submission and verification process, ensuring a smooth and user-friendly experience.

2. OCR and Text Extraction Module:

The OCR module extracts text from uploaded documents such as certificates, marksheets, and scorecards, processing scanned, handwritten, or low-quality images into machine-readable text. Pre-processing techniques like noise removal, contrast enhancement, and text normalization are used to improve extraction accuracy and ensure reliable data retrieval from various document formats.

3. Translation and Language Normalization Module:

This module converts extracted text from regional or multilingual documents into English using a Translation API, ensuring uniformity during the verification process. It resolves language discrepancies and supports documents from diverse linguistic backgrounds, enabling seamless multilingual handling within the system.

4. NLP-Based Verification Module:

The NLP module analyzes extracted text using tokenization, entity recognition, and semantic matching to validate key fields such as name, date of birth, certificate number, and category. It automatically detects inconsistencies or mismatches between user-entered biodata and extracted document information, improving accuracy and reducing manual verification effort.

5. Distributed Processing (MapReduce) Module:

This module ensures high-speed document verification by distributing OCR and NLP tasks across multiple computation nodes using a MapReduce model. It enables the system to verify each document within seconds, supporting large-scale recruitment workflows efficiently and reliably.

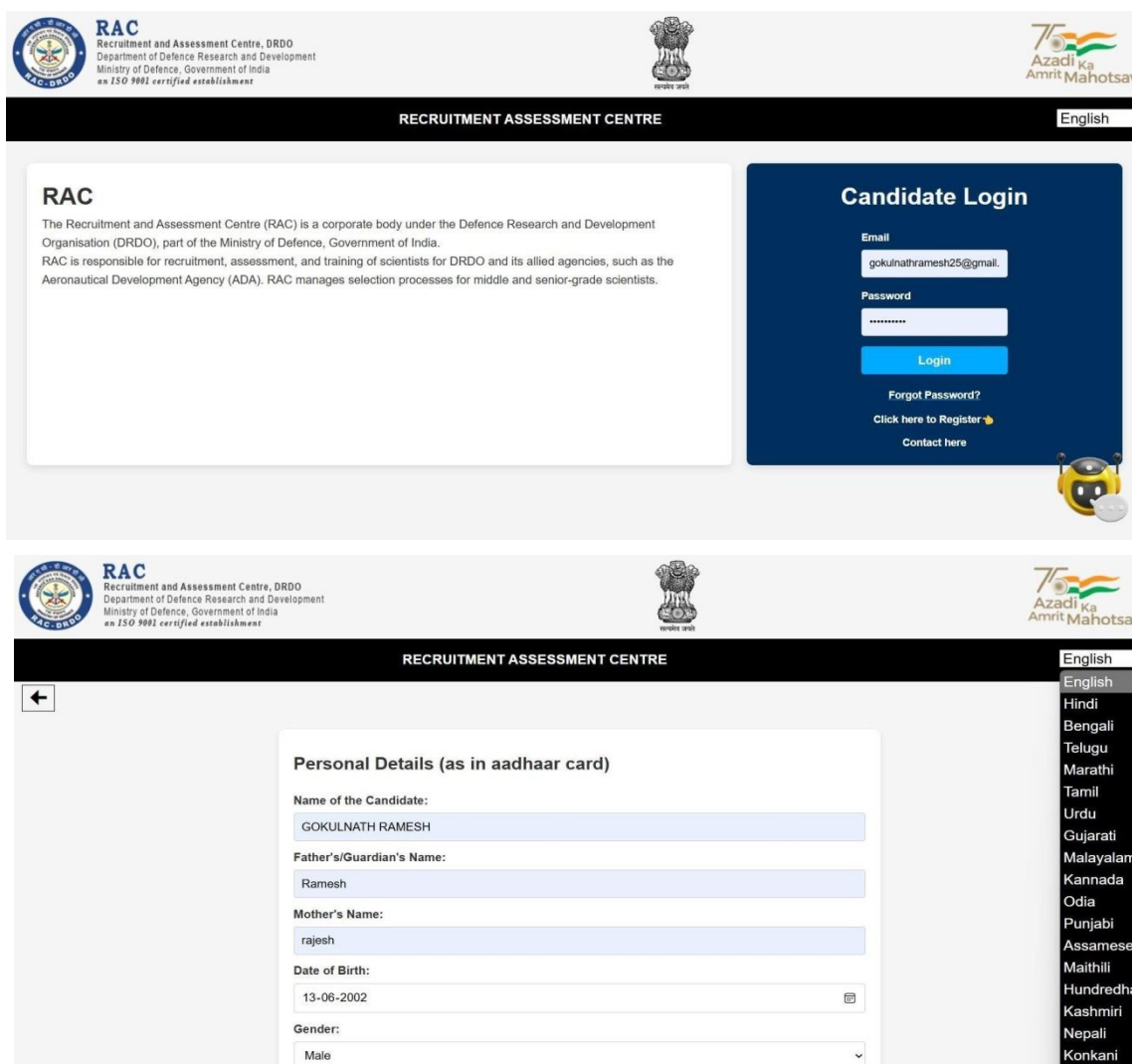
6. Alert and Notification Module:

This module generates immediate alerts for candidates and recruiters when mismatches or anomalies are detected during verification. It provides clear error messages and recommendations, allowing candidates to correct and resubmit data while enabling recruiters to quickly review flagged issues.

7. Secure Storage and Database Module:

The database module securely stores uploaded documents, extracted text, verification results, and final validated records using encrypted storage mechanisms. It ensures data integrity, confidentiality, and easy retrieval for future auditing, reporting, or recruitment decision-making.

VI. EXPECTED OUTCOME



The screenshot displays the Recruitment Assessment Centre (RAC) website. The top header includes the RAC logo, the text "Recruitment and Assessment Centre, DRDO", "Department of Defence Research and Development", "Ministry of Defence, Government of India", and "an ISO 9001 certified establishment". The central navigation bar reads "RECRUITMENT ASSESSMENT CENTRE" with a language dropdown set to "English".

The main content area is divided into two sections. On the left, under the heading "RAC", there is a brief description: "The Recruitment and Assessment Centre (RAC) is a corporate body under the Defence Research and Development Organisation (DRDO), part of the Ministry of Defence, Government of India. RAC is responsible for recruitment, assessment, and training of scientists for DRDO and its allied agencies, such as the Aeronautical Development Agency (ADA). RAC manages selection processes for middle and senior-grade scientists." On the right, the "Candidate Login" section features input fields for "Email" (gokulnathramesh25@gmail.com) and "Password" (masked with dots), a "Login" button, and links for "Forgot Password?", "Click here to Register", and "Contact here".

Below the login section, a "Personal Details (as in aadhaar card)" form is visible. It includes input fields for "Name of the Candidate:" (GOKULNATH RAMESH), "Father's/Guardian's Name:" (Ramesh), "Mother's Name:" (rajesh), "Date of Birth:" (13-06-2002), and "Gender:" (Male). A language dropdown menu is open on the right side of the form, listing various Indian languages including English, Hindi, Bengali, Telugu, Marathi, Tamil, Urdu, Gujarati, Malayalam, Kannada, Odia, Punjabi, Assamese, Maithili, Hundredha, Kashmiri, Nepali, and Konkani.

VII. CONCLUSION

In conclusion, the development of an AI-driven Intelligent Document Processing system for automated verification marks a significant advancement in modern recruitment workflows. By integrating OCR for accurate text extraction, NLP for intelligent comparison, and translation support for multilingual documents, the system effectively addresses the limitations of traditional manual verification. The use of distributed processing models ensures fast and scalable performance, enabling documents to be verified within seconds while maintaining high reliability. This automated approach reduces human error, enhances transparency, and supports fair decision-making in recruitment processes. Ultimately, the system empowers organizations to handle large-scale applications efficiently and provides candidates with a smoother, more trustworthy experience. By promoting accuracy, security, and operational efficiency, this solution contributes to a more advanced, digitized, and equitable recruitment ecosystem.

VIII. REFERENCE

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